

MAPÚA UNIVERSITY

Final Project Documentation

Employee Management System for Barangay Hulo Office

In partial fulfillment for the course

Web Systems and Technologies 2 Laboratory

IT135-8L

Submitted by:

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Submitted to:

Prof. Geldof Resuello

May 2023

Introduction

Among the twenty-seven barangays in the City of Mandaluyong, Barangay Hulo is situated in the southern vicinity of the city acting as one of the boundaries between the metropolis of both Makati and Mandaluyong. The barangay covers an area measurement of approximately 20+ hectares and is home to a population estimated to be around 30,000 residents as of 2020.

Barangay Hulo local government office is accountable for supplying its constituents with a variety of essential services, such as public health and safety, waste management, medical services, and community development/social programs. It is led by barangay captain *Bernardino C. Maglaque* and a number of barangay councilors who are elected by the barangay's citizens. Moreover, the following local government unit's office is responsible for overseeing the management system for over 100 employees working under the barangay. The elements of the system include the payroll process and 'application for leave services' for every staff member. However, there is no existing employee management system yet for the barangay's LGU office as any employee related transactions are done through the usually adapted paper-based manual filling process. Pay slips and file for leave forms are filled through manual handwriting.

Problem Statement & Objectives

At this current time, there is no existing employee management system, whether manual or automated, in the LGU's office. Therefore, tracking and managing each employee will be an inefficient and time-consuming approach. With that being said, it can cause delayed payroll and leave requests to be problematic. To address this problem, the group aims to automate the system by building a web system that caters for both office administrators' and employee's convenience. The proposed web system will streamline the entire employee management procedure, allowing

employees to digitally submit leave requests and management to conveniently approve them. Furthermore, the platform will also automate the payroll process.

With this, the group aims to accomplish the following objectives:

1. Create a web system that allows both admin and user to access unique functionalities.
2. Automate employee management system for administrator.
3. Automate employee pay slip retrieval and application for leave.

UN SDGs Addressed

Specifically, the proposed project addresses the following United Nation Sustainable Development Goals:

1. **Goal 8: Promote sustained, inclusive and sustainable economic growth, full and productive employment and decent work for all –** the proposed web system contributes to the conceptual achievement of SDG goal 8 by positively improving the working conditions of employees through the utilization of an automated employee management system that fosters productivity. The following project will surely reduce the time and effort needed for manual administrative processes. This allows both the management and staff to focus more on other productive activities that will return economic growth for their local office.

2. **Goal 9: Build resilient infrastructure, promote inclusive and sustainable industrialization and foster innovation** – the proposed web system directly addresses goal 9 as it utilizes the innovative approach of web technologies to automate and streamline the process of employee management systems. This encourages the innovative growth of technological infrastructure since it enables the LGU office to invest in the software and equipment needed for system implementation and integration. Overall, this technological investment could lead to the expansion of jobs and the enhancement of

employees' technical skills, which then contributes to the achievement of inclusive and sustainable industrialization.

Scope of the Project

The project will be focused on creating an employee management web application. It will include two account types, employee, and admin. The platform is mostly used by the administrator since an employee account only has limited features in the application. The web system will only entertain employees in the LGU office, the admin of the website can add new employees, and delete/edit existing ones.

Functional/ Non-functional requirements

Functional Requirements

User Authentication: The system should have a user authentication mechanism that allows employees and administrators to log in with their credentials.

Employee Management: The administrator can add new employees, as well as edit and delete existing employee records. The admin can view different employees assigned depending on each department.

Department Management: The admin should have the ability to add, edit, and delete departments.

Leave Management: The system should allow employees to create leave requests. At the same time, the admin should be able to see the leave requests made by different employees, as well as approve or reject those requests.

Payroll Management: The system should allow employees to view their payroll history.

Payment Approval: The admin should be able to view the amount to be paid for each employee, as well as make the transfer to them.

Non-functional Requirements

Usability: Both administrator and employees should find the system easy to use and navigate.

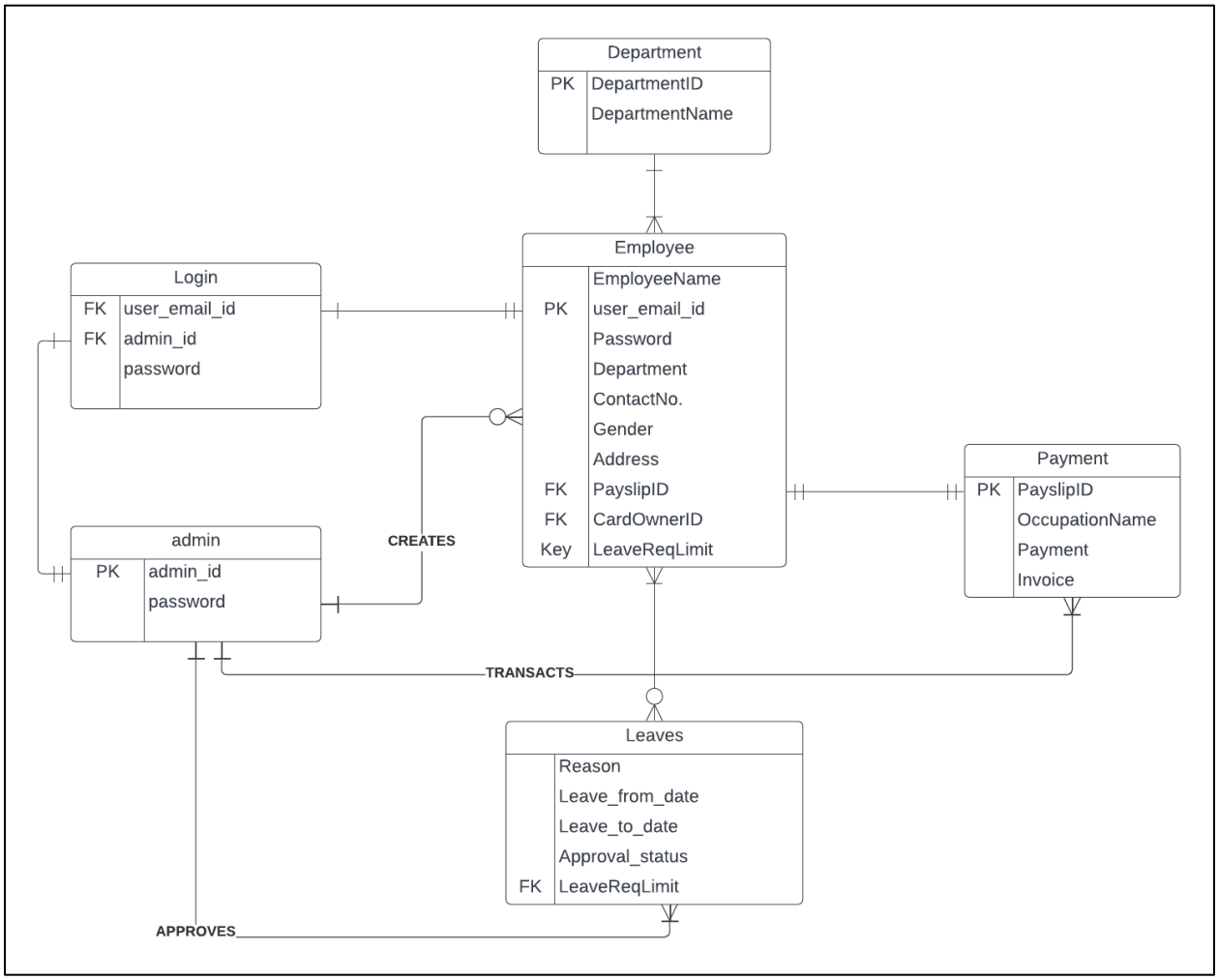
Reliability: There should be little to no downtime and constant availability of the system.

Performance: There should be no performance issues when handling a high volume of users and requests.

Security: To protect the employee's data from unauthorized access, data breaches, and other security threats, the system should have effective security measures in place.

Compatibility: To guarantee that it can be used by all users, the system should be compatible with a range of browsers and hardware.

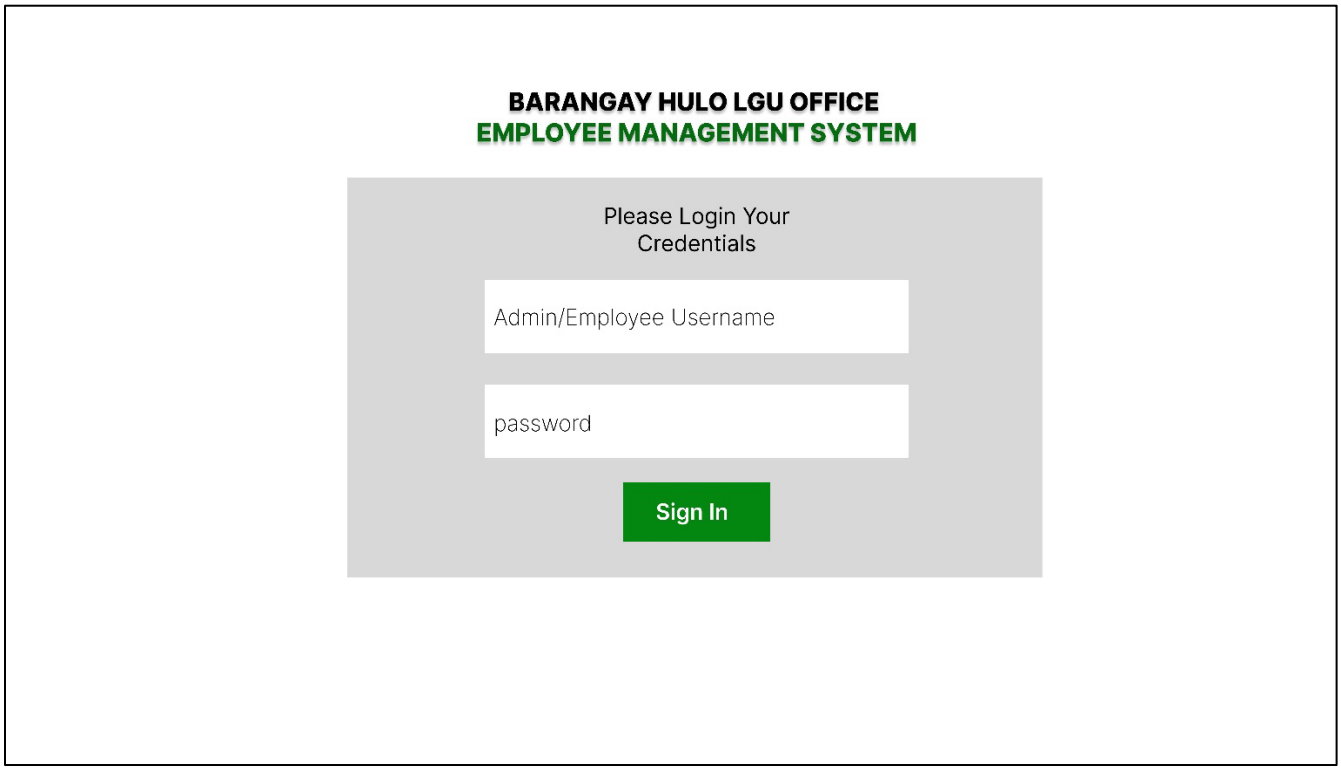
Database design (ERD)



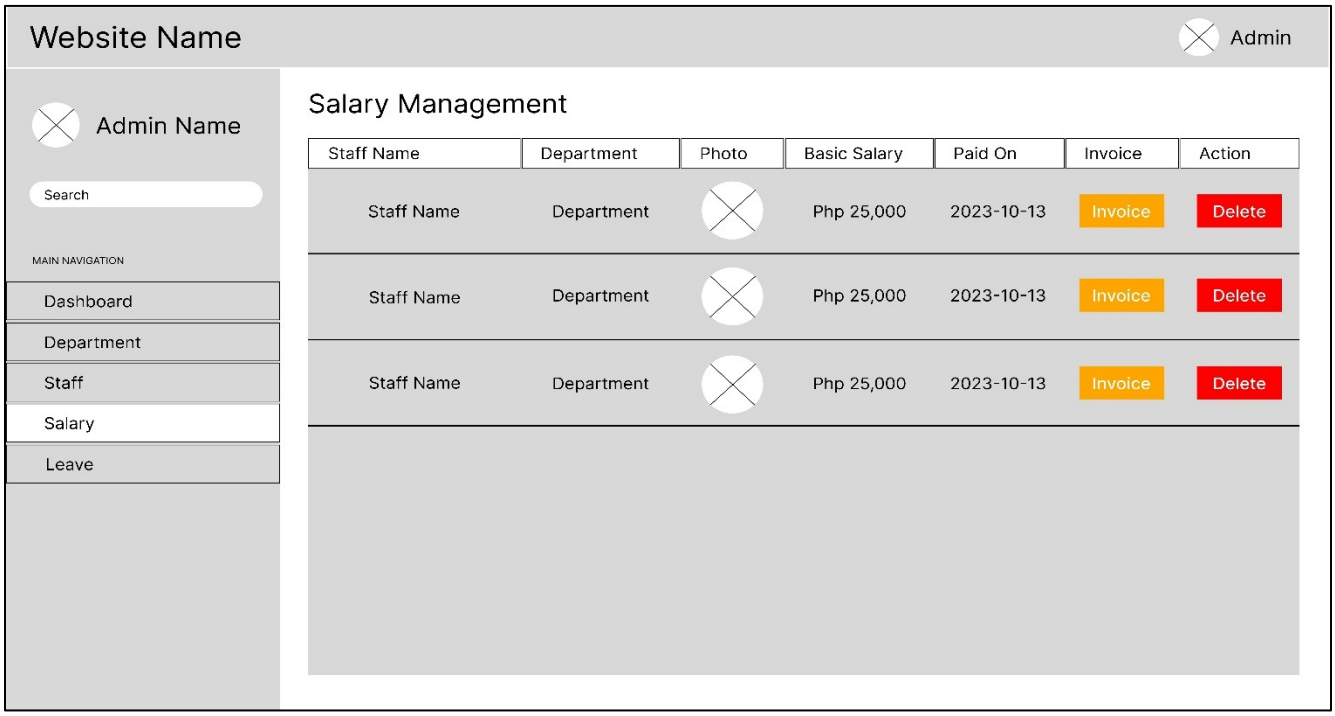
The following entity relationship diagram describes the overall process of the proposed project “Employee Management System for Barangay Hulo Office”. To start, the management system has a single login page for both employee and login. Login entity illustrates the attributes that are required for both admin and employee login action. Department entity shows a “one or many” relationship with table Employees, while the opposite Employees to Department is a “one to one” relationship. This is because the department function could have many employees, but the employee should only have a single department. Admin entity then creates the Employees, transacts the Payments, and approves the Leaves for the employees. It has a “zero to many” relationship with Employees because the admin could create 100+ employees, and a “one to many” relationship with both Payment and Leave entities since it will only process either a single or numerous payments and leave approval/denial depending on the number of employees who submitted their requests. All in all, the relationship of these other entities to the admin is a “one to one” because the assumption is that there is a single administrator managing the system.

Features of the System

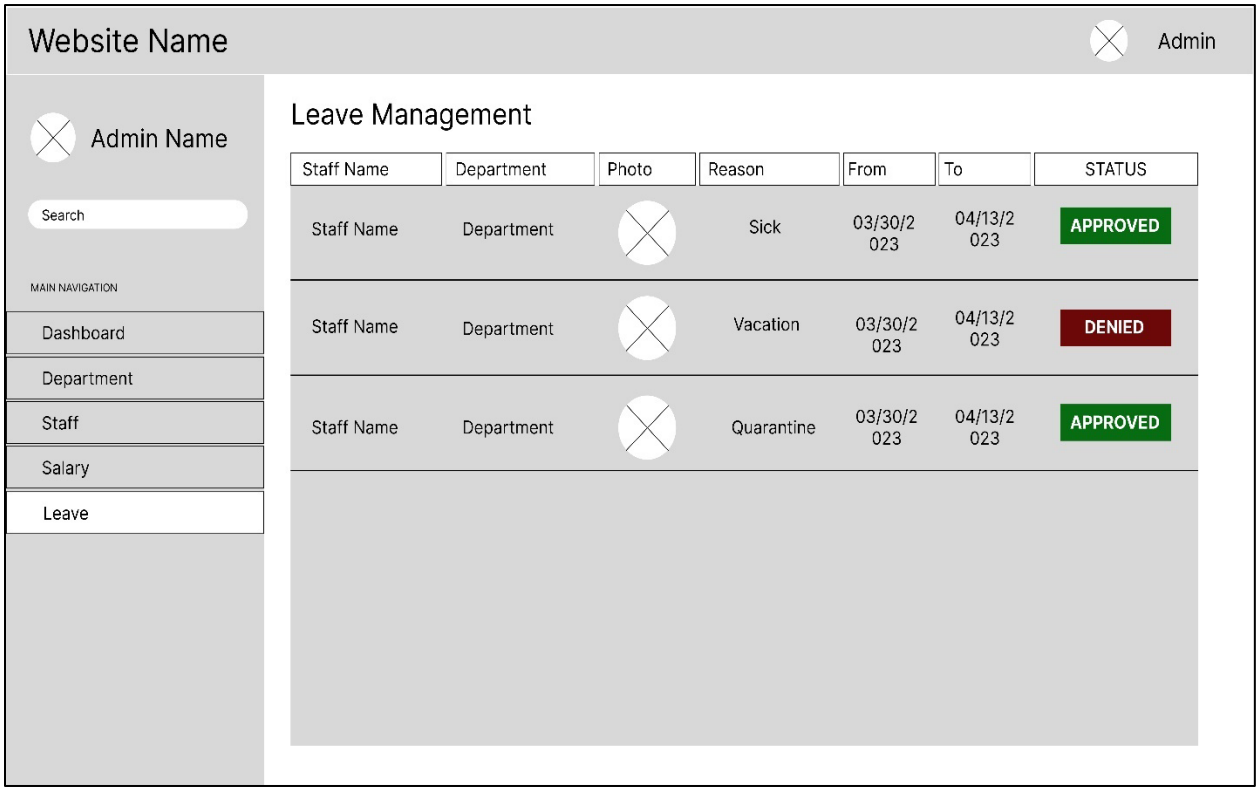
Admin/Employee Login:



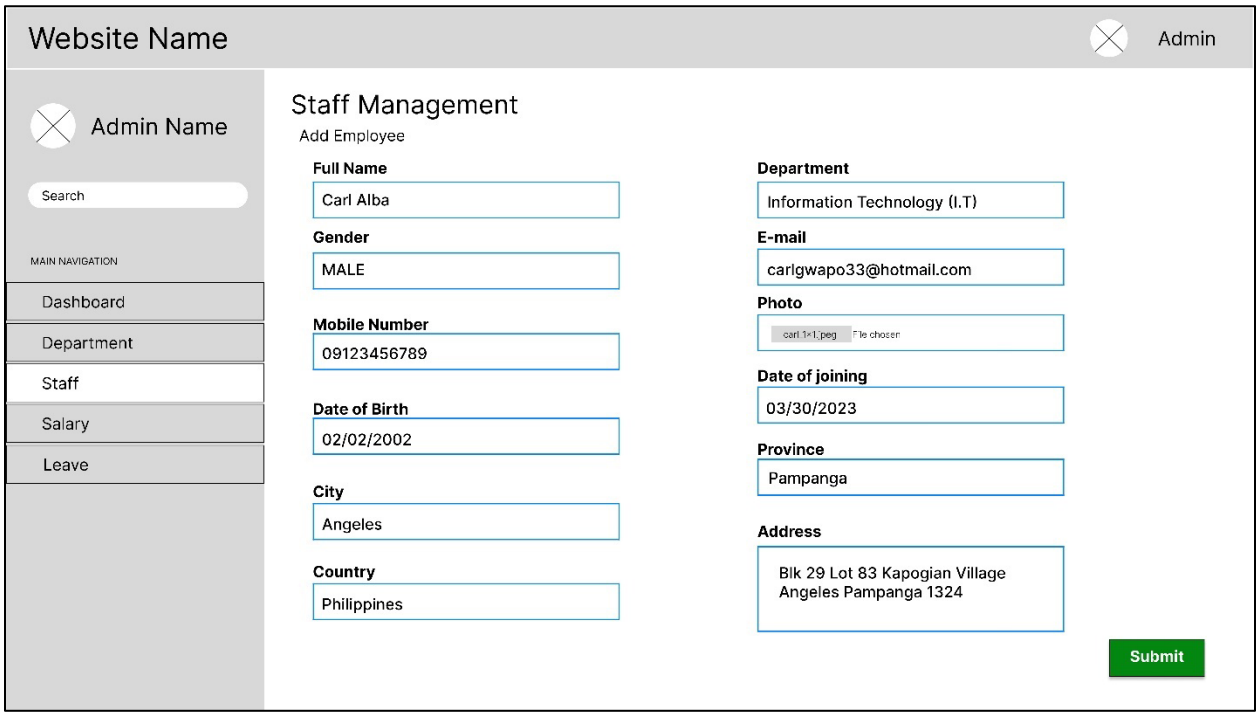
Admin: Salary Management Frame



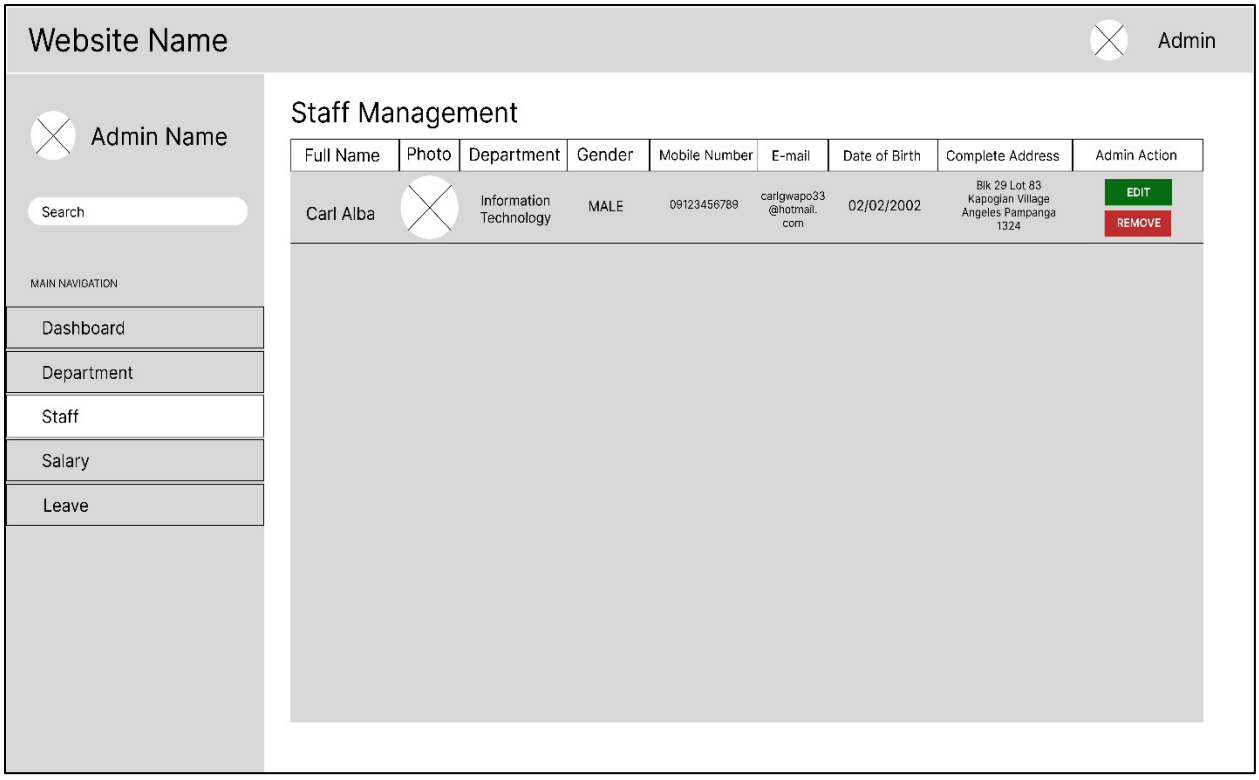
Admin: Leave Management Frame



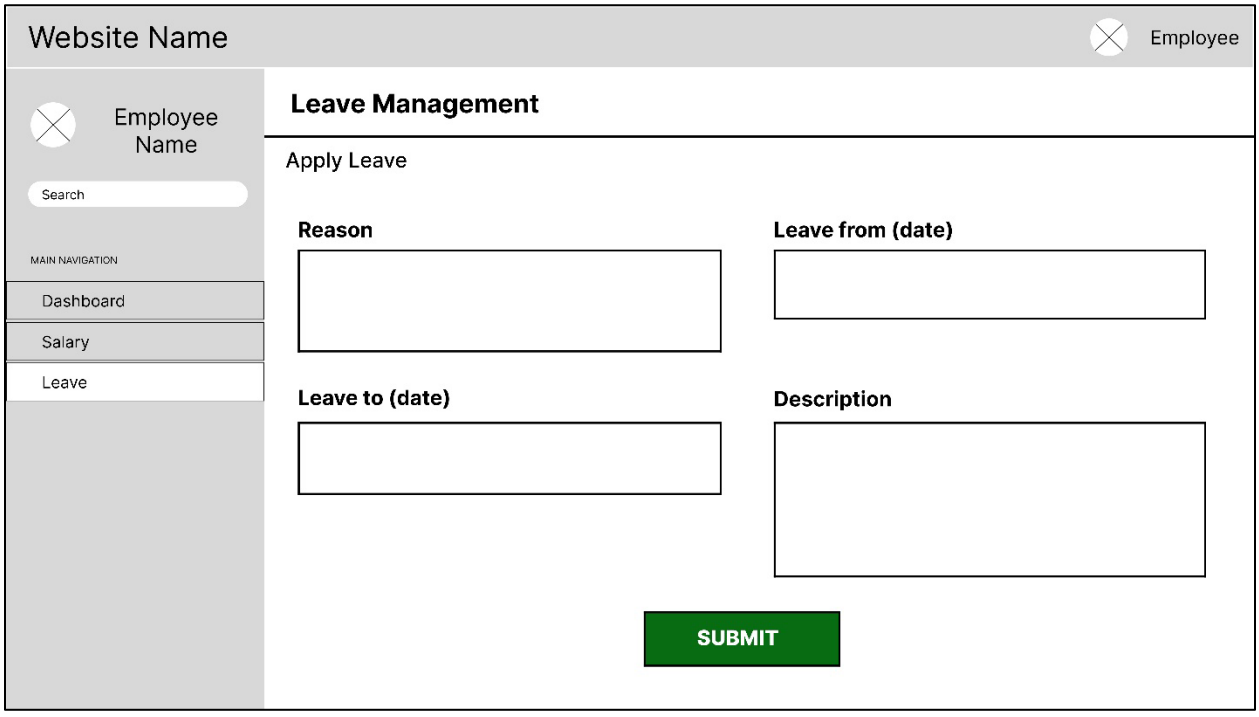
Admin: Staff Management – Add Employee



Admin: Staff Management – Edit or Remove Employee



Employee: Leave Management – Application of Leave



The said application will be including features such as:

For administrators:

- Adding new employee to the list
- Editing/Deleting existing employees
- Adding/Deleting/Editing of department
- Viewing of existing employees in the department
- Approving the payment for the employees
- Approval of leave requests from the employees.

For employees:

- Viewing the payroll history
- Filing leave requests

Communication Letter with the client of choice

28 March 2023

BERNARDINO C. MAGLAQUE
Barangay Captain, Hulo

Dear Mr. Bernardino C. Maglaque,

I hope this letter finds you in good health and high spirits. As a major requirement for our subject, Web Systems and Technologies 2, we, the third-year Bachelor of Science majoring in Computer Science students, are writing to discuss creating an employee management system for Barangay Hulo.

As you may know, effective employee management is crucial for the success of any organization. In Barangay Hulo, where there are numerous employees across various departments, it is essential to have an efficient system in place to manage their schedules, organize payroll, and ensure that they are meeting their goals.

We propose creating an employee management system that will streamline employee management. The system will be web-based, accessible to all department heads and administrators, and allow for easy attendance and performance evaluation monitoring.We have already begun researching different systems and technologies suitable for your needs. We can create a system that will significantly improve your employee management processes with your support and cooperation.

Thank you for your attention to this matter, and we look forward to discussing this further with you.

Respectfully yours,



CARL JOSEPH S. ALBA
Group Leader

WEBSITE LINK: https://barangayhulo.000webhostapp.com/

TEST RESULTS

Test Scenario	Login for both employee and admin			
STEP Num#	*Step Details*	*Expected Results*	*Actual Results*	*Status*
1	Access login page	Website should open	As expected	Pass
2	Input username/email and password	User can type into the fields	As expected	Pass
3	Click "Sign In" button	Admin/Staff is logged in	As expected	Pass

Test Scenario	Log out for both employee and admin			
STEP Num#	*Step Details*	*Expected Results*	*Actual Results*	*Status*
1	Click upper right profile button	Profile pane should pop out	As expected	Pass
2	Click "Sign Out" button	Admin/user will be logged out.	As expected	Pass

Test Scenario	Employee View: Side navigation functions			
STEP Num#	*Step Details*	*Expected Results*	*Actual Results*	*Status*
1	Click "Dashboard" button	User will be redirected to dashboard pane	As expected	Pass
2	Click "Salary" button	Button expand as "View Salary" option will pop out.	As expected	Pass
3	Click "View Salary" button	It will show salary management view	As expected	Pass
4	Click "Leave" button	Button expand as "Apply Leave" and "My Leave History" option pop out	As expected	Pass

5	Click “Apply Leave”	Redirects employee to “Leave Management” form.	As expected	Pass
6	Click “My Leave History”	Redirects employee to “Leave Management” history.	As expected	Pass

Test Scenario	Employee View: Application for leave form			
STEP Num#	*Step Details*	*Expected Results*	*Actual Results*	*Status*
1	Click “Leave” button	Button expand showing “Apply Leave” option.	As expected	Pass
2	Click “Apply leave” button	User is redirected to apply leave form	As expected	Pass
3	Type in “Reason” field	User can type into the field	As expected	Pass
4	Input in “Leave from date” field	User can input into the field	As expected	Pass
5	Input in “Leave to” date	User can input into the field	As expected	Pass
6	Type in “description” field	User can type into the field	As expected	Pass
7	Click “Submit” button	Form is submitted with success confirmation. The application is forwarded to the admin and the user’s my leave history	As Expected	Pass

Test Scenario	Employee View: Salary Management			
STEP Num#	*Step Details*	*Expected Results*	*Actual Results*	*Status*
1	Click “Salary” button	Button expand showing “View Salary” option.	As expected	Pass
2	Click “View Salary”	User is redirected to view salary records	As expected	Pass
3	Click “Invoice” button	User is redirected to the invoice information pane	As expected	Pass

4	Click “Print” button	Prompt of printable version of invoice will be displayed	As expected	Pass
5	Click “Generate PDF” button	Prompt of PDF version of invoice will be displayed	As expected	Pass
6	Click “Drop down” entries	Will filter data based on selected entries	As expected	Pass
7	Click “previous, next, and page counters”	Can navigate through different pages	As expected	Pass

Test Scenario	Admin View: Dashboard functionalities			
STEP Num#	*Step Details*	*Expected Results*	*Actual Results*	*Status*
1	Click “Dashboard” button	Redirect admin to dashboard page	As expected	Pass
2	Click “Departments” from control panel dashboard	Redirect admin to Manage Departments page	As expected	Pass
3	Click “Staff” from control panel dashboard	Redirect admin to Manage Staff page	As expected	Pass
4	Click “Leave Requests” from control panel dashboard	Redirect admin to Leave Management page	As expected	Pass
5	Click “Salary Paid” from control panel dashboard	Redirect admin to Salary Management page	As expected	Pass

Test Scenario	Admin View: Department functionalities			
STEP Num#	*Step Details*	*Expected Results*	*Actual Results*	*Status*
1	Click “Department” button	Button expands showing other Department options	As expected	Pass
2	Click “Add Department” in expanded options.	Redirect admin to Add Department page	As expected	Pass
3	Type in “Department Name” field	Allow admin to type in department name	As expected	Pass

4	Click “Submit” in add department	It will save the added “department” to the DB and Manage Department page.	As expected	Pass
5	Click “Manage Departments” in the expanded options	Redirect admin to Management Departments page and display Departments record	As expected	Pass
6	Delete one department from the record	It should delete the chosen department from the Manage Departments record.	As expected	Pass
7	Edit one department from the record	It should allow the admin to edit the Manage Departments record.	As expected	Pass

Test Scenario	Admin View: Staff functionalities			
STEP Num#	*Step Details*	*Expected Results*	*Actual Results*	*Status*
1	Click “Staff” button	Button expands showing other Staff options	As expected	Pass
2	Click “Add Staff” in expanded options.	Redirect admin to Add Staff page	As expected	Pass
3	Type in “Add Staff” fields	Allow admin to type in Staff personal information	As expected	Pass
4	Click “Submit” in add staff management	It will save the added “staff” to the DB and Manage Staff page.	As expected	Pass
5	Click “Manage Staff” in the expanded options	Redirect admin to Management Staff page and display Staff records	As expected	Pass
6	Delete one staff from the record	It should delete the chosen staff from the Manage staff record.	As expected	Pass
7	Edit one staff from the record	It should allow the admin to edit the Manage Staff record.	As expected	Pass

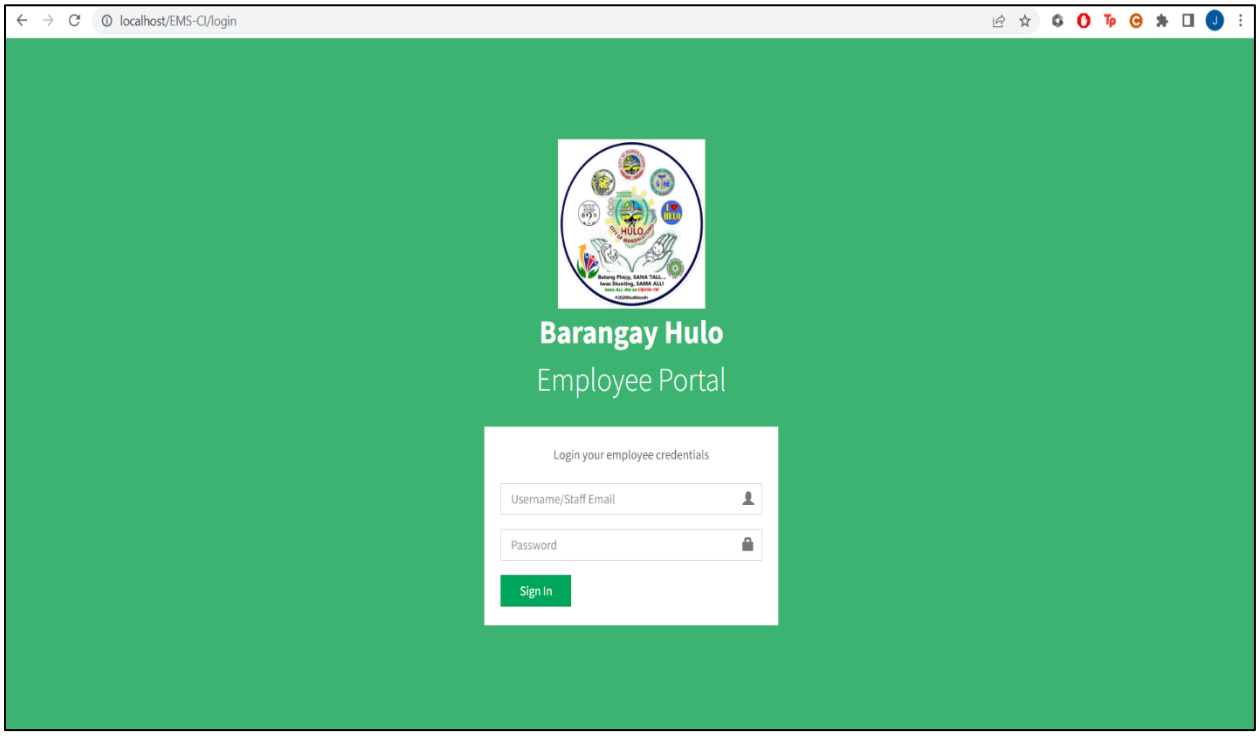
Test Scenario	Admin View: Salary functionalities			
STEP Num#	*Step Details*	*Expected Results*	*Actual Results*	*Status*
1	Click “Salary” button	Button expands showing other salary options	As expected	Pass
2	Click “Add Salary” in expanded options.	Redirect admin to Add Salary page	As expected	Pass
3	Choose department name	Allow admin to choose department name in Add Salary page	As expected	Pass
4	Input Basic Salary and Allowance values	Allow admin to input salary values	As expected	Pass
5	Click “Submit” button.	It will save the added “salary” to the DB and Manage Salary page.	As expected	Pass
6	Click “Manage Salary” option	Redirects admin to the Salary Management page	As expected	Pass
7	Delete one salary from the record	It should delete the chosen salary from the Manage salary record.	As expected	Pass
8	Click “Invoice” button	It should display invoice details	As expected	Pass
9	Click “Print” button	Prompt of printable version of invoice will be displayed	As expected	Pass
10	Click “Generate PDF” button	Prompt of PDF version of invoice will be displayed	As expected	Pass
11	Click “Drop down” entries	Will filter data based on selected entries	As expected	Pass
12	Click “previous, next, and page counters”	Can navigate through different pages	As expected	Pass

Test Scenario	Admin View: Leave functionalities			
STEP Num#	*Step Details*	*Expected Results*	*Actual Results*	*Status*
1	Click “Leave” button	Button expands showing other Leave options	As expected	Pass

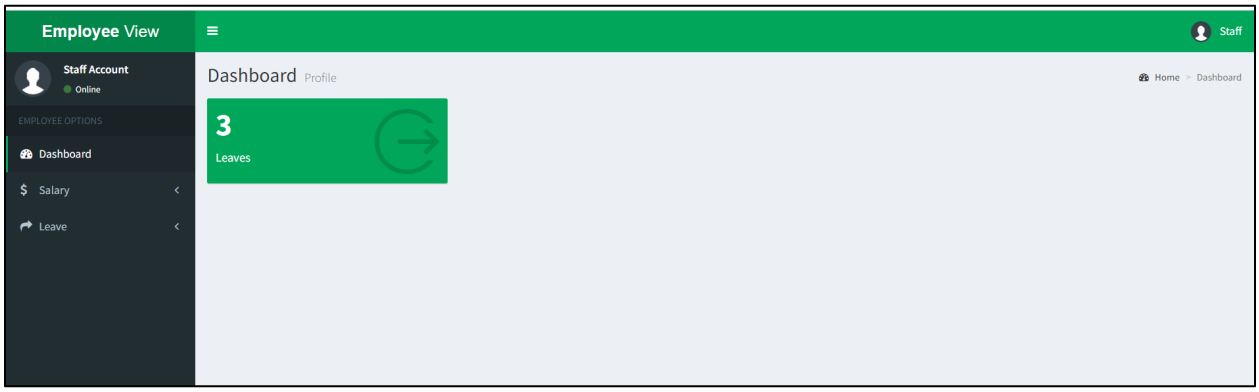
2	Click “Manage Staff’s Leave” in expanded options.	Redirect admin to Leave Management page showing leave application records	As expected	Pass
3	Click “Approve” button	Allow admin to approve application and save it in Leave historypage	As expected	Pass
4	Click “Reject” button	Allow admin to reject the application and save it in Leave history page	As expected	Pass
5	Click “Leave History” in the expanded options	Redirect admin to Leave History page and display leave application historical records	As expected	Pass

FUNCTIONS OF THE SYSTEM (SCREENSHOTS)

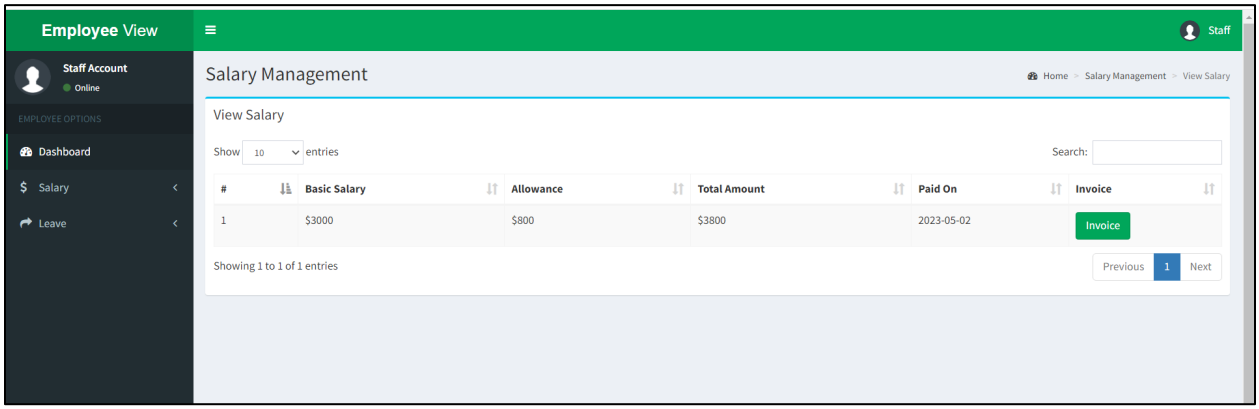
EMPLOYEE/ADMINISTRATOR LOGIN PAGE:



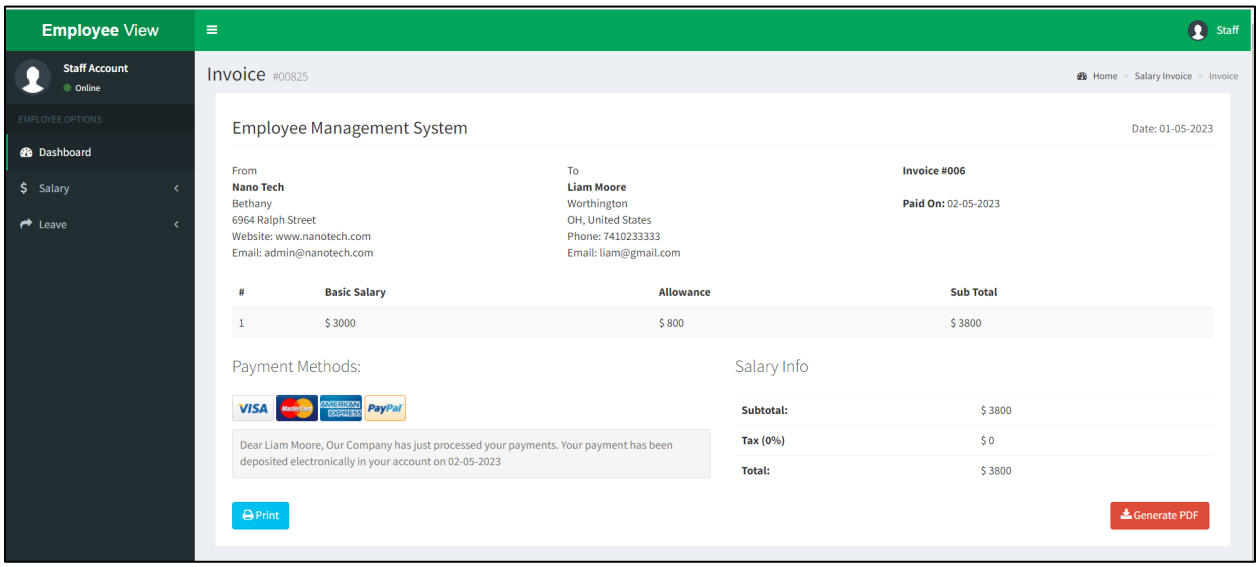
Employee: Dashboard View



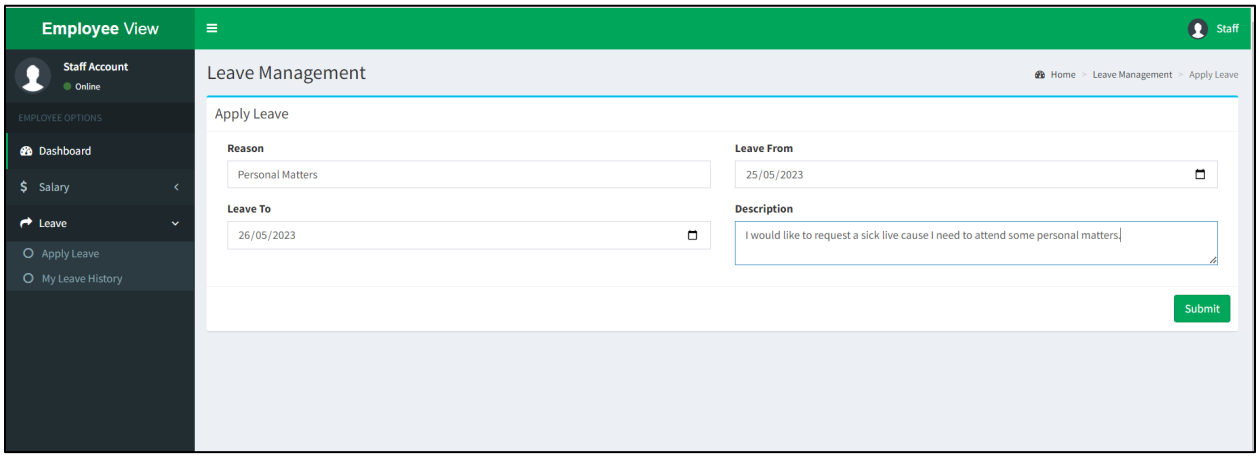
Employee: View Salary



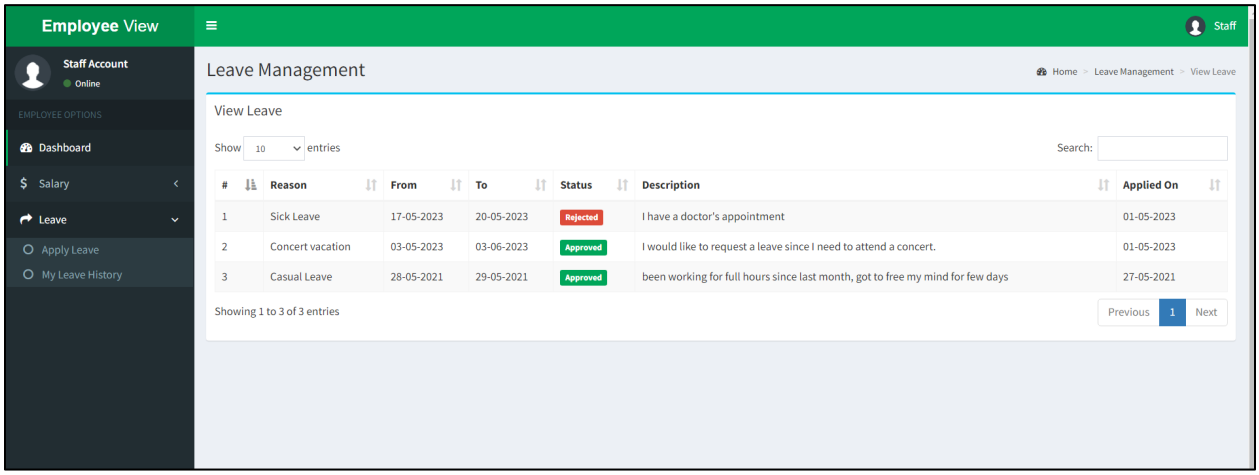
Employee: View Invoice



Employee: Leave Application



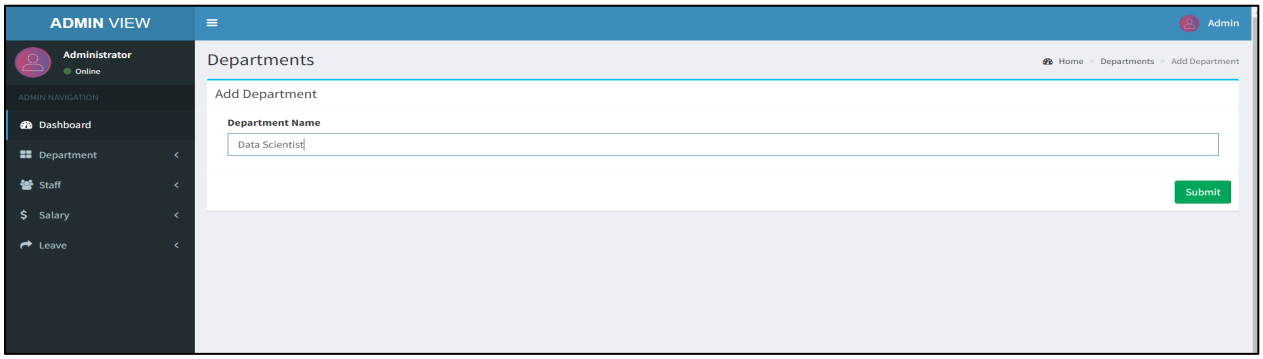
Employee: My Leave History



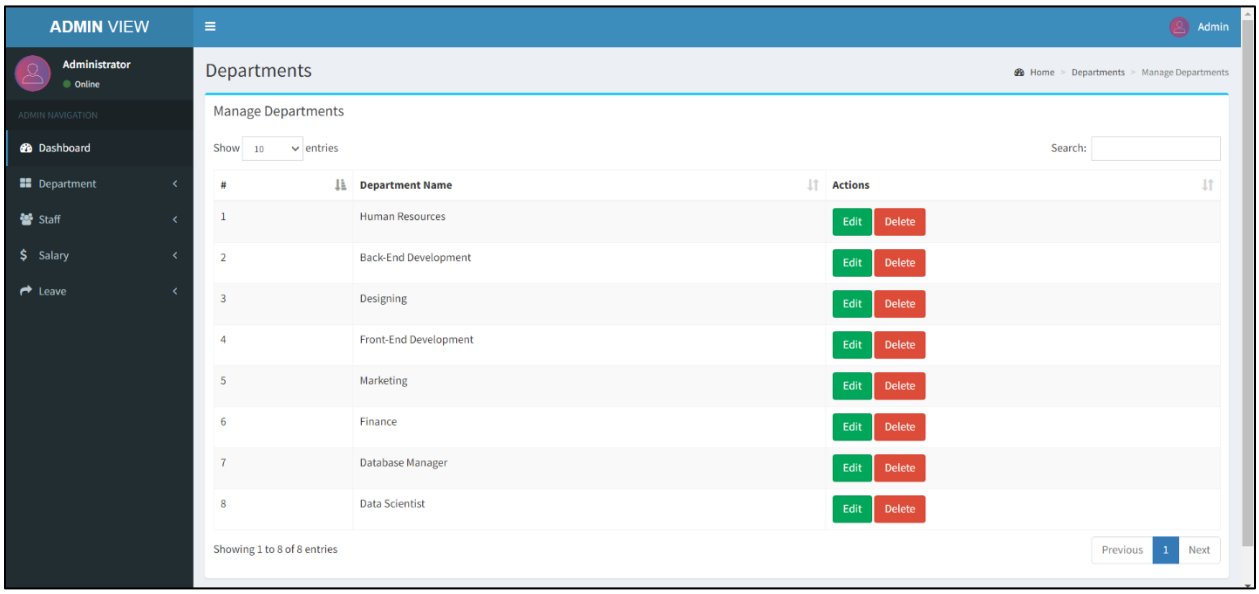
ADMIN: Dashboard View



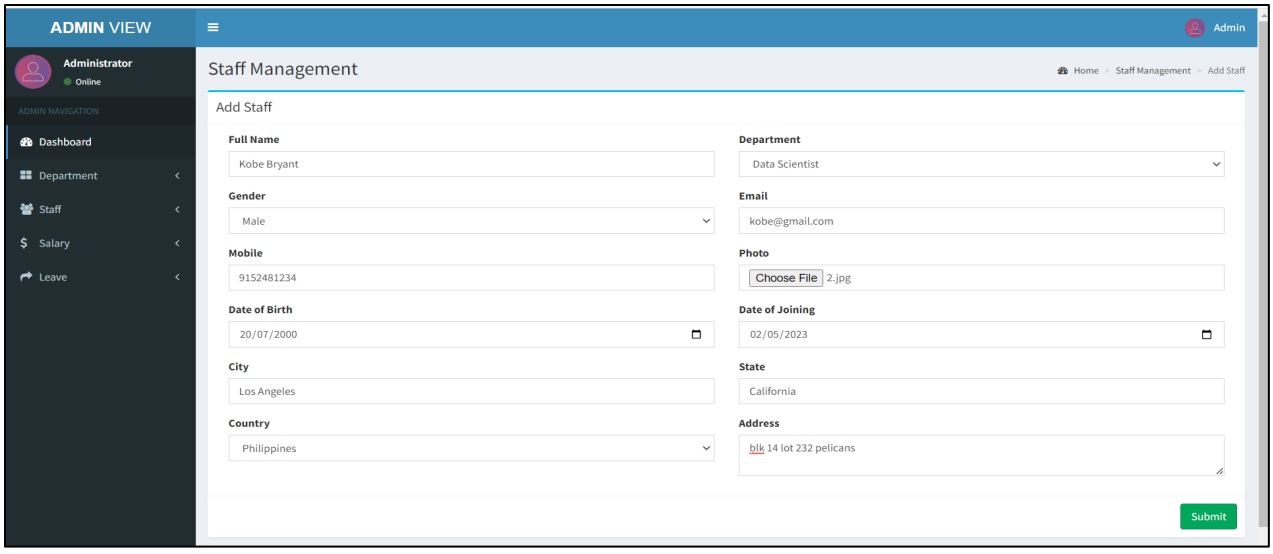
ADMIN: Add Department



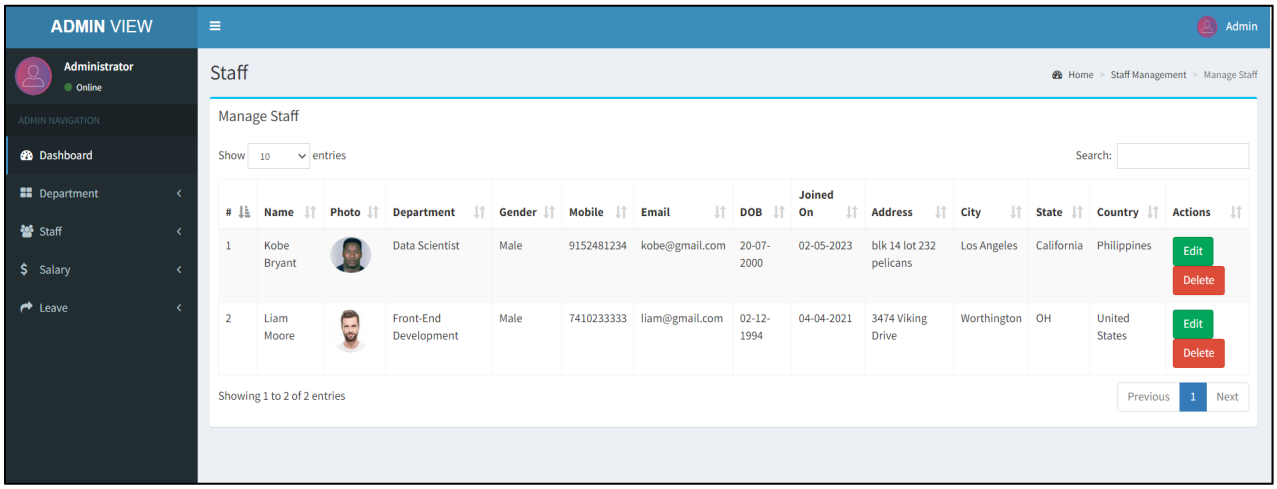
ADMIN: Manage Departments



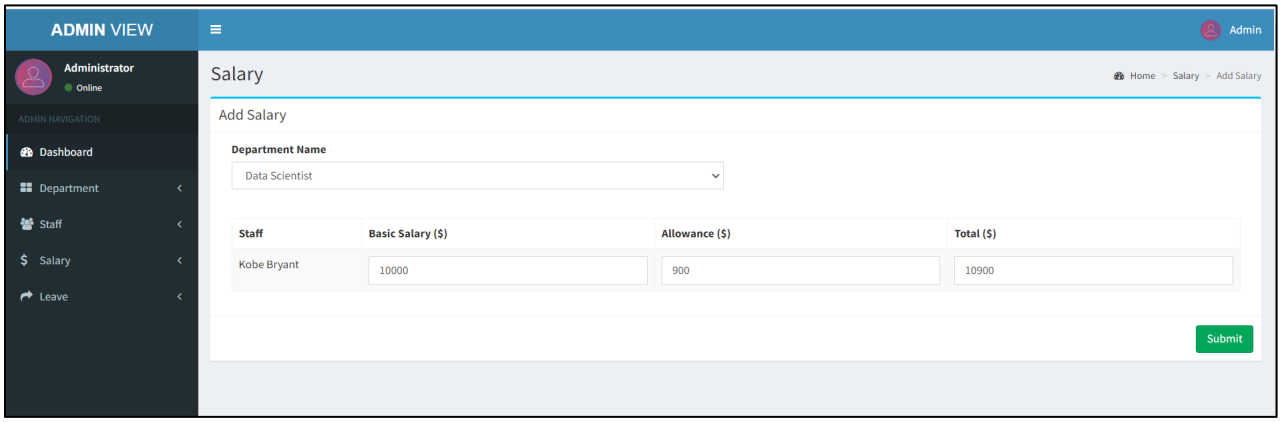
ADMIN: Add Staff



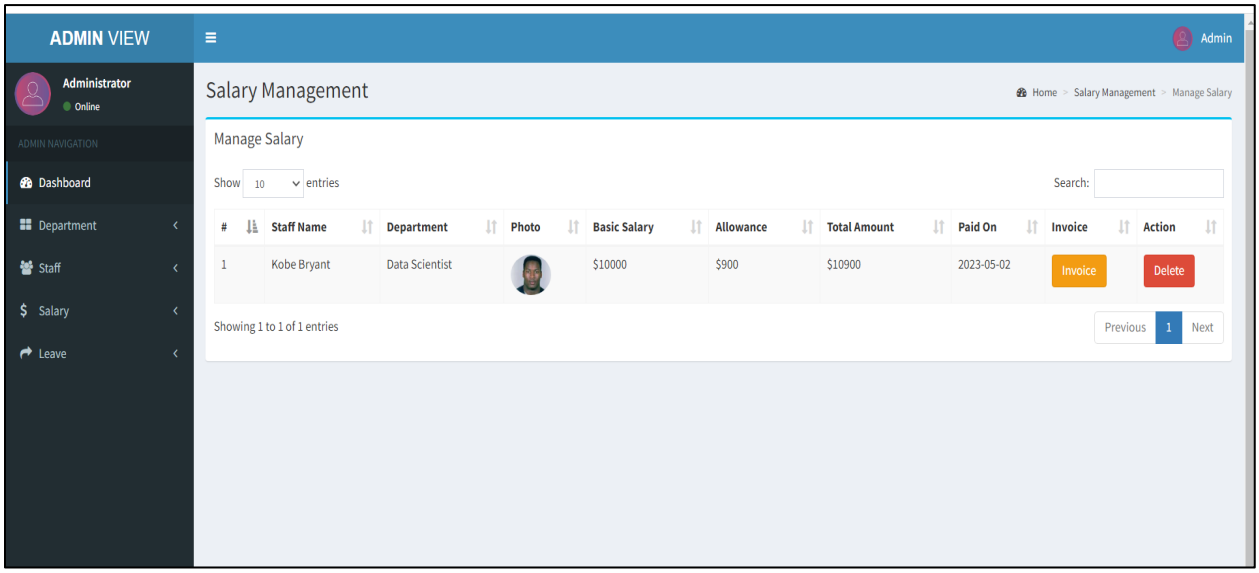
ADMIN: Manage Staff



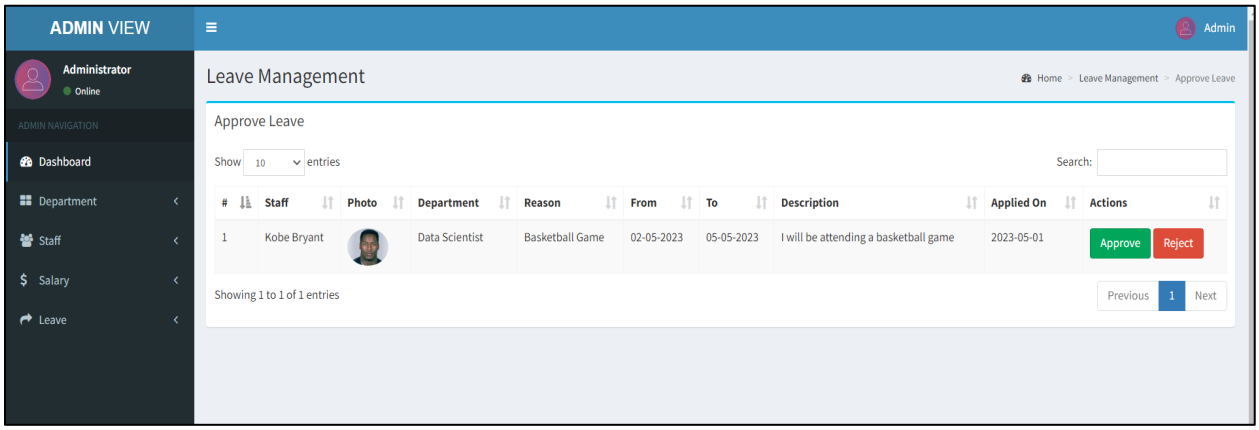
ADMIN: Add Salary



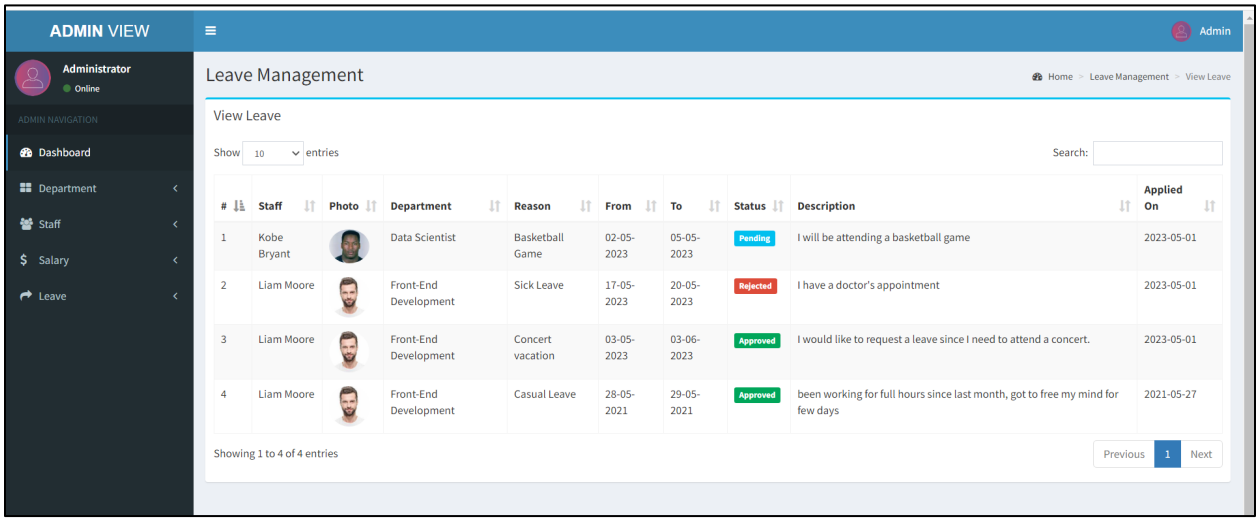
ADMIN: Salary Management



ADMIN: Leave Management



ADMIN: Leave History



PROJECT SIGN-OFF

Barangay Hulo
Employee Portal

Authorization Memorandum

I have carefully assessed the specifications and deliverables for the *Barangay Hulo Employee Management System.*

MANAGEMENT CERTIFICATION - Please check the appropriate statement.

_/_ The project deliverables are accepted.

We would like to take this opportunity to express our appreciation for your support and cooperation throughout the project. We are grateful for the opportunity to work with you and look forward to future collaborations.

By signing this document, we fully accept the finalized changes and deliverables as final. Based on our authority and judgment, the continued operation of this system is authorized.

Carl Joseph Alba

John Kyle Bautista

Joshua Reynald Labrado

Adrian Joseph Maglaque

PROJECT MEMBERS

Bernardino C. Maglaque
Brgy. Chairman of Partnered LGU Barangay Hulo

Professor Cristina Pascua
PROJECT MENTOR